ACOS HW2

no1. (6-bit)

Unsigned:

13 = 1101

6-bit → 001101

24 = 11000

6-bit → 011000

63 = 111111 ← 6-bit

Signed:

16 = 10000

6+bit → 010000

2 = 000010 => -2 = 111101 and adding 1 = 111110 ← 6bit

31 = 11111

6-bit → 011111

-32 = 100000 ← 6bit

no2.

Unsigned:

000101

$1 \cdot 2^0 + 1 \cdot 2^2 = 5$

101011

$1 \cdot 2^0 + 1 \cdot 2^1 + 1 \cdot 2^3 + 1 \cdot 2^5 = 43$

111111

$1 \cdot 2^0 + 1 \cdot 2^1 + 1 \cdot 2^2 + 1 \cdot 2^3 + 1 \cdot 2^4 + 1 \cdot 2^5 = 63$

100000

$1 \cdot 2^5 = 32$

Signed:

Significant bit = 0 => positive

000101 = 5

Significant bit = 1 => negative

invert and add 1 = 010101

$-1 \cdot 2^0 + (-1) \cdot 2^2 + (-1) \cdot 2^4 = -21$

111111 = -1

Significant bit = 1 => negative

invert and add 1 = 100000 = -32

№ 3.

7 = 111

8-bit → 00000111

Hexadecimal = 07, because 0000 = 0 and 0111 = 7

240 = 11110000 ← 8-bit

Hexadecimal = F0

171 = 10101011 ← 8-bit

Hexadecimal = AB

126 = 01111110 ← 8-bit

Hexadecimal = 7E

№ 4.

0x3C = 00111100, because 3 = 0011

0x7E = 01111110

0xFF = 11111111

0xA5 = 10100101

№ 5.

Invert and add 1:

0x3C = 00111100 = (inv. + 1) = 11000100

0x7E = 01111110 = (inv. + 1) = 10000010

0xFF = 11111111 = (inv. + 1) = 00000001

0xA5 = 10100101 = (inv. + 1) = 01011011

No 6.

0x DEADBEEF

Big-Endian:

MSB in the highest and LSB in the lowest:

DE AB BE EF

Little-Endian:

LSB in the highest and MSB in the lowest:

EF BE AD DE

No 7.

7 = 00 111 Sign-exten. = 00000 111 ~~Zer~~ 0-exten. = 00000 111

8-bit → 000 00 111

15 = 0 1111

Sign-exten. = 0000 1111

Zero-exten = 0000 1111

-16 = 00000

Sign-exten. = 00000000

zero-exten. = 00000000

-5 = 0 1011

[redacted]

Sign-exten. = 000 01011

zero-exten. = 000 01011

No 8.

UnSigned:

7 = 0111

9 = 1001

0111 + 1001 = 10000

Signed:

4 = 0100 0100 + 0101 = 1111 or -1

-5 = 0101